Alexander Brown

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PROFESSIONAL SUMMARY

Georgia Tech Computer Science M.S. candidate with 2+ years building web applications for 9,000+ annual users. Experience in React/TypeScript, Python APIs, SQL, automated testing, and ML.

EDUCATION

Georgia Institute of Technology August 2025 - Expected May 2027
M.S. Computer Science, Artificial Intelligence Specialization
Relevant Coursework: Machine Learning, Deep Learning, Computer Networks, Health Informatics
GPA: 4.0

University of California, Irvine September 2022 - June 2025
B.S. Computer Science, Intelligent Systems Specialization
Relevant Coursework: Artificial Intelligence, Databases, Systems, Data Structures, Algorithm Analysis, Information Retrieval, Embedded Systems, Graph Algorithms, Probability, Statistics
GPA: 3.85

RESEARCH

Graduate Research Assistant May 2026 - September 2026
Georgia Tech Human Augmented Analytics Group (HAAG)
- Benchmarked YOLO, RT-DETR, DETR, and Swin detectors on Himalayan camera-trap footage.
- Ran HPC ablations across hyperparameters, augmentation, and data splits for annotation-limited detection.

Machine Learning Research Assistant September 2024 - April 2025
California Institute for Telecommunications and Information Technology (Calit2), UC Irvine
- Developed PyTorch GRU, LSTM, RNN, and RBF-RNN models for thermal energy storage, achieving sub-second inference versus a 10-minute traditional simulation.

WORK EXPERIENCE

Full-Stack Web Developer March 2023 - June 2025
University of California, Irvine, Student Affairs Information Technology
- Developed and maintained Laravel/PHP, Vue.js, and MySQL applications supporting 9,000+ annual users across UCI student-facing systems.
- Automated end-to-end QA workflows with Cypress and custom scripts to improve release reliability.

PROJECTS

Medical Bill and EOB Assistant October 2026 - Present
Full-Stack Developer | React, TypeScript, Django, PostgreSQL | In progress
- Developing upload and extraction with pdfplumber and Tesseract OCR, converting bills and EOBs into structured JSON for an LLM-assisted review workflow.
- Implementing Python checks for duplicate charges and arithmetic discrepancies; contributing to the shared chat UI and component integration.

Probabilistic Weather Downscaling October 2026 - Present
Data Pipeline and Diffusion Developer | Python, PyTorch, Zarr | In progress
- Building ERA5/CONUS404 ingestion, 8-km regridding, normalization, and data loaders for California weather downscaling.
- Developing conditional diffusion and ensemble-sampling components that connect the data pipeline with model training and evaluation.

AI Body-Worn Camera Footage Tagging January 2025 - June 2025
Team Lead | 1st Place, Computer Science, UCI ICS Expo
- Built an Electron/React review app and FastAPI backend for Signal Hill Police Department's 16-TB footage-tagging initiative.
- Integrated Faster-Whisper transcription, visual-similarity models, and ensemble classification into the footage review workflow.

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, PHP, Java, C++
Web and Data: React, Vue.js, Django, Laravel, FastAPI, PostgreSQL, MySQL
Tools and ML: Git, Docker, AWS, Bash, Cypress, PyTorch, scikit-learn, Claude Code, Codex